

ELEMENTS OF AERONAUTICAL ENGINEERING LAB

Semester								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6AE01	ESC	L	T	P	C	CIA	SEE	Total
		-	-	3	2	40	60	100
UNIT-I	Basics of Aerodynamics							
Introduction-Anatomy of airships, gliders, airplane, helicopter, UAV - Forces on Flight Vehicles - Lift generation on the airplane - Drags and Types of Drag, Estimation of forces and moments on airfoil- Introduction to airfoils- nomenclature, Types of Airfoils- NACA Series and their applications, Laminar flow airfoils, Low Reynolds number airfoils								
UNIT-II	Basics of Aerospace structures							
Airship structural components- glider structural components, Airplane structural components-wing components- functions of each component-fuselage components-tail components and functions- landing gear components and functions, Helicopter structural components- rotor blades -tail rotors and functions								
UNIT-III	Aerospace Propulsion systems							
Introduction -principles of thrust generation- Thrust equation - propulsion used in airships reciprocating engine parts-propellers-gas turbine engine parts-functions-demonstration- engine starters- Introduction to rocket engines								
UNIT-IV	Aircraft systems							
Introduction-hydraulic system components and functions- pneumatic systems- components and functions- electrical system-components-cabin-pressurization system-anti-icing and deicing systems-landing gear system-cockpit display systems-Instruments-helicopter display system-instruments								
UNIT-V	Evolving technologies in Aeronautical Engineering							
Introduction-advanced control systems-components and functions, Applications of AI/MLI-Product Designing-Better Fuel Efficiency-Effective Supply Chain Management-Air Traffic Management-Identification of Threats- Applications of data science-Weather patterns and maintenance schedules. Data is collected, from flight takeoff to landing.Data patterns - analysis and algorithms for a safer flight Advanced flight deck display systems								
Text Books:								
1. Newman, D., Interactive Aerospace Engineering and Design, (with software and reference material on CD),McGraw-Hill, 2002, ISBN 0-07-112254-0. 2. Anderson, J.D., Introduction to Flight, fifth edition, Tata McGraw-Hill, 2007, ISBN: 0-07-006082-								
Reference Books:								
1. Barnard, R. H. and Philpot, D.R., Aircraft Flight, 3rd edition,Pearson, 2004,ISBN: 81-297-0783- 2. Machine Learning For Absolute Beginners by Oliver Theobald, 2020 3. A C Kermode. Flight Without Formulae, 5 th edition Pearson Education.								

2008,ISBN:9788131713891

Course Outcomes : Students should be able to:

1. Distinguish the evolution of aerofoil profile and to describe the history of Aeronautical Engineering
2. Analyze how lift, drag and thrust are generated and understand which components constitute them
3. Analyze the flight performance parameters with respective stability condition
4. Distinguish the different systems used in an aircraft
5. Analyze the new technologies used in Aeronautical Engineering